

Digital Electronics

Number System

Decimal <u>Base = 10</u>	Binary <u>Base = 2</u>	Octal <u>Base = 8</u>	Hexa-decimal <u>Base = 16</u>
0	0	0	0 to 9
1	1	1	10 - A
2		2	11 - B
3		3	12 - C
4		4	13 - D
5		5	14 - E
6		6	15 - F
7		7	
8			
9			

Any Base to Decimal [Conversion]

$$(\overset{2}{P} \overset{1}{Q} \overset{0}{R} . \overset{-1}{X} \overset{-2}{Y} \overset{-3}{Z})_8$$

$$= (P \times 8^2 + Q \times 8^1 + R \times 8^0 + X \times 8^{-1} + Y \times 8^{-2} + Z \times 8^{-3})_{10}$$

Binary to Decimal

$$\begin{aligned}(1010)_2 &= ()_{10} \Rightarrow (1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0)_{10} \\ &= (8 + 0 + 2 + 0)_{10} \\ &= (10)_{10}.\end{aligned}$$

$$\begin{aligned}(101010)_2 &= ()_{10} \Rightarrow (1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + \\ &\quad 0 \times 2^0)_{10} \\ &= (32 + 8 + 2)_{10} \\ &= (42)_{10}.\end{aligned}$$

Octal to Decimal

$$\begin{aligned}(237.456)_8 &= ()_{10} \Rightarrow (2 \times 8^2 + 3 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1} + 5 \times 8^{-2} \\ &\quad + 6 \times 8^{-3})_{10} \\ &= (159.589)_{10}.\end{aligned}$$

Hexa-decimal to Decimal

$$\begin{aligned}(A2DC.9FED)_{16} &= ()_{10} \\ &= (A \times 16^3 + 2 \times 16^2 + D \times 16^1 + C \times 16^0 + 9 \times 16^{-1} \\ &\quad + F \times 16^{-2} + E \times 16^{-3} + D \times 16^{-4})_{10} \\ &= (41692.6245)_{10}.\end{aligned}$$

Decimal to Binary (Sum of Weighted Method)

$$(10)_{10} = (?)_2, (12)_{10} = (?)_2, (14)_{10} = (?)_2.$$

	8	4	2	1
10 -	1	0	1	0
12 -	1	1	0	0
14 -	1	1	1	0

Binary to Octal

To convert binary to octal, we have to make group of three digits (moving left from point for the digits on the left side of point and moving right from point for the digits on the right side of the point). And write its equivalent octal.

$$(10110110101110.01011001000101010110)_2 = (?)_8$$

Left Side

110 → 6
101 → 5
110 → 6
110 → 6
010 → 2

Right Side

010 → 2
110 → 6
010 → 2
001 → 1
010 → 2
101 → 5
100 → 4

$$\Rightarrow (26656.26212540)_8.$$

Octal to Binary

$$(65332.26)_8 = (110101011011010.010110)_2.$$

Binary to Hexa-decimal

$$(01011011011011)_2 = (?)_{16}$$

$$1011 \rightarrow 11 (B)$$

$$1101 \rightarrow 15 (F)$$

$$0110 \rightarrow 6$$

$$0001 \rightarrow 1$$

$$(01011011011011)_2 = (16FB)_{16}.$$

Hexadecimal to Binary

$$(\bullet 16FB)_{16} = (0001011011011011)_2.$$

Octal to Hexadecimal

$$(65\cancel{1}3)_8 = (?)_{16}$$

$$(65\cancel{1}3)_8 = (110101011)_2$$

$$(000\cancel{1}110101011)_2 = (1AB)_{16}$$

$$(653)_8 = (1AB)_{16}.$$

Hexadecimal to Octal

Hexadecimal $\rightarrow$ Binary $\rightarrow$ Octal.

Binary Addition

$$\begin{array}{r} \overset{\textcircled{0}}{1} \overset{\textcircled{0}}{0} 1 1 \\ + 1 0 0 1 \\ \hline 1 0 1 0 0 \end{array}$$

$$\begin{array}{r} \overset{\textcircled{1}}{1} \overset{\textcircled{0}}{1} 0 1 \\ + 1 1 0 1 \\ \hline 1 1 0 1 0 \end{array}$$

1's Complement

$$\neg(1 0 0 1) = 0 1 1 0$$

$$1's \text{ complement of } 1101 = 0010$$

2's Complement

$$\boxed{2's \text{ complement} = 1's \text{ complement} + 1}$$

Q. 1010

$$\begin{array}{r} 1's \text{ complement} = 0101, \quad \overset{\textcircled{1}}{0} 1 0 1 \\ + 0 0 0 1 \\ \hline 0 1 1 0 \end{array}$$

$$2's \text{ complement of } 1010 = 0110.$$

Q. 10101101

$$\begin{array}{r} 1's \text{ complement} = 01010010, \quad 0 1 0 1 0 0 1 0 \\ + 0 0 0 0 0 0 0 1 \\ \hline 0 1 0 1 0 0 1 1 \end{array}$$

$$2's \text{ complement of } 10101101 = 01010011$$

Binary Subtraction Using 1's Complement

$$(10)_{10} - (8)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 1000 \xrightarrow{1's\ comp} 0111 \end{array} \longrightarrow \begin{array}{r} \overset{0}{1} \overset{0}{0} 10 \\ + 0111 \\ \hline \boxed{1} 0001 \\ 0001 \\ + 0001 \\ \hline 0010 \end{array}$$

$$(10)_{10} - (8)_{10} = (0010)_2 = (2)_{10}.$$

Binary Subtraction Using 2's Complement

$$(10)_{10} - (8)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 1000 \xrightarrow{2's\ comp} 1000 \end{array} \longrightarrow \begin{array}{r} 1010 \\ + 1000 \\ \hline \cancel{X} \boxed{1} 0010 \end{array}$$

$$(10)_{10} - (8)_{10} = (2)_{10}.$$

$$(10)_{10} - (2)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 0010 \xrightarrow{2's\ comp} 1110 \end{array} \longrightarrow \begin{array}{r} \overset{0}{1} \overset{0}{0} 10 \\ + 1110 \\ \hline \cancel{X} \boxed{1} 1000 \end{array}$$

$$(10)_{10} - (2)_{10} = (8)_{10}.$$

DeMorgan's Theorem

There are two theorems:-

1. Complement of a product is equal to sum of complements.

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

$$\overline{A \cdot B \cdot C} = \bar{A} + \bar{B} + \bar{C}$$

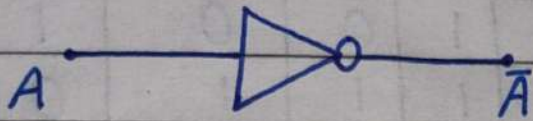
2. Complement of a sum is equal to product of complements.

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

$$\overline{A + B + C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

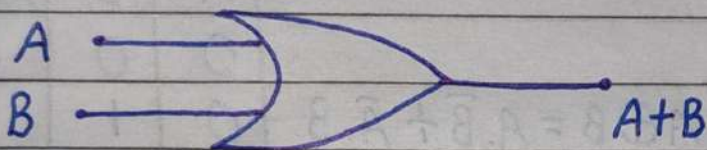
Basic/Standard Gates

NOT



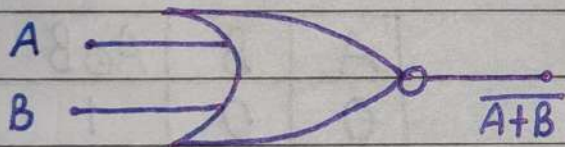
A	$\bar{A}$
0	1
1	0

OR



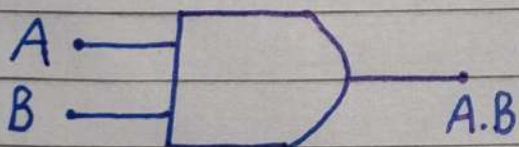
A	B	$A+B$
0	0	0
0	1	1
1	0	1
1	1	1

NOR



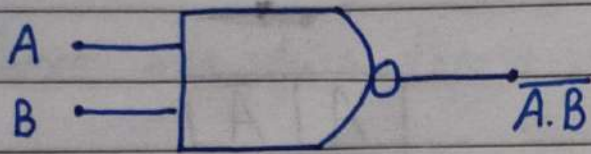
A	B	$A+B$	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

AND



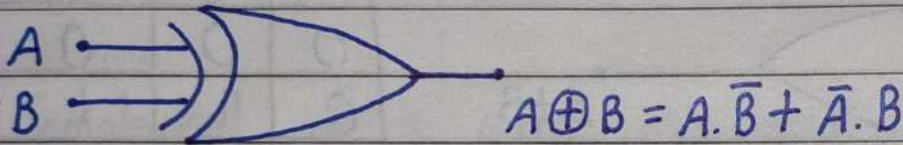
A	B	$A.B$
0	0	0
0	1	0
1	0	0
1	1	1

NAND



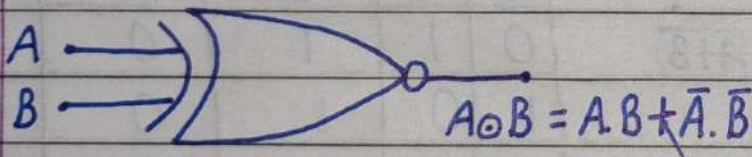
A	B	$A.B$	$\overline{A.B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Imp: XOR (Exclusive OR)



A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Imp: X-NOR (Exclusive NOR)



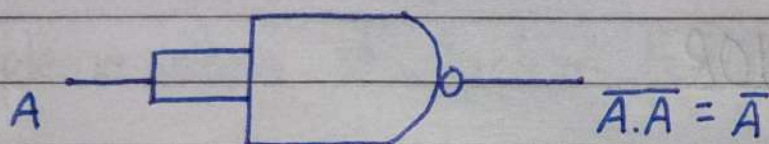
A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

Imp # Universal Gates

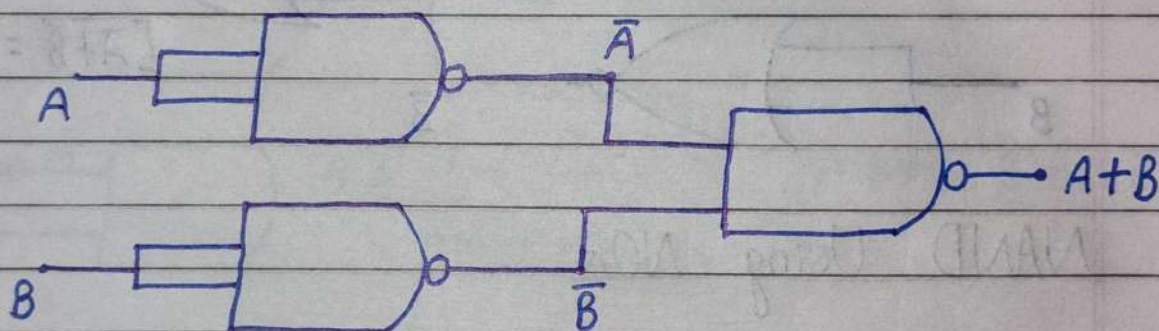
NAND & NOR gates are called as universal gates because any boolean function can be implemented/realized using these (NAND/NOR) gates.

Note: $A.A.A = A$ & $A+A+A = A$.

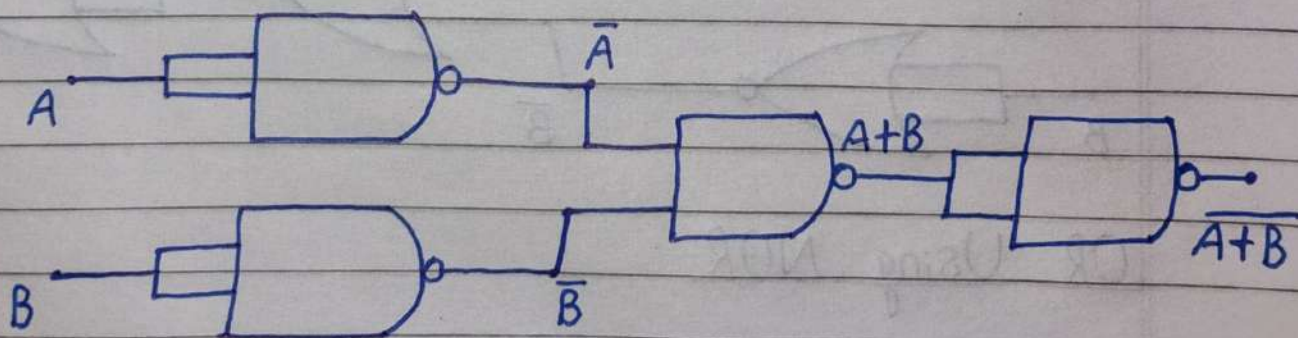
NOT Using NAND



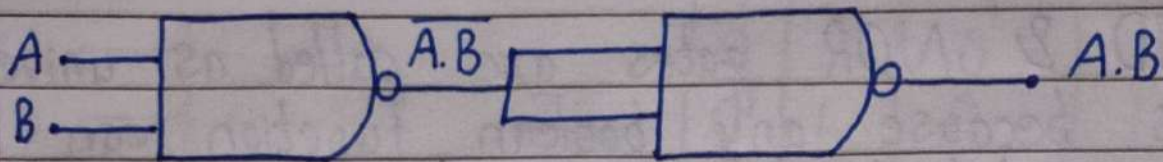
OR Using NAND



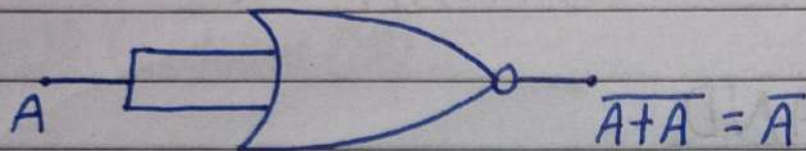
NOR Using NAND



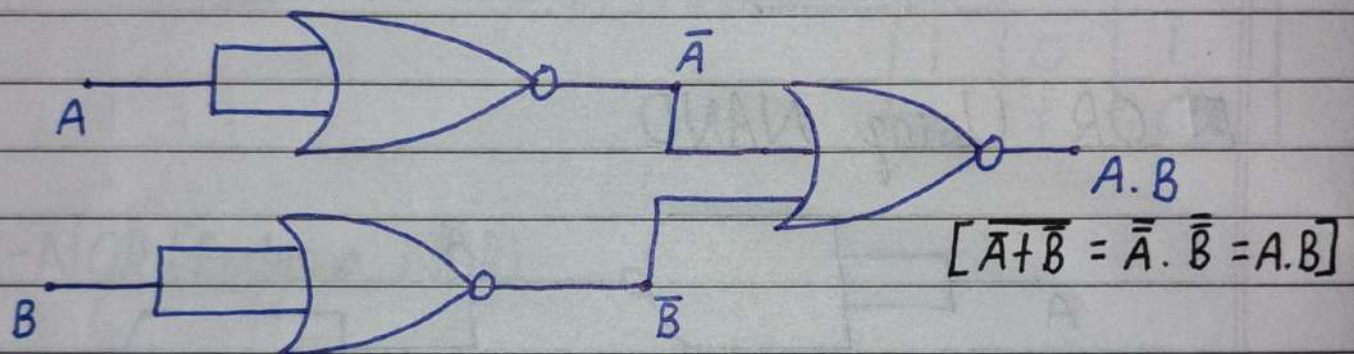
AND Using NAND



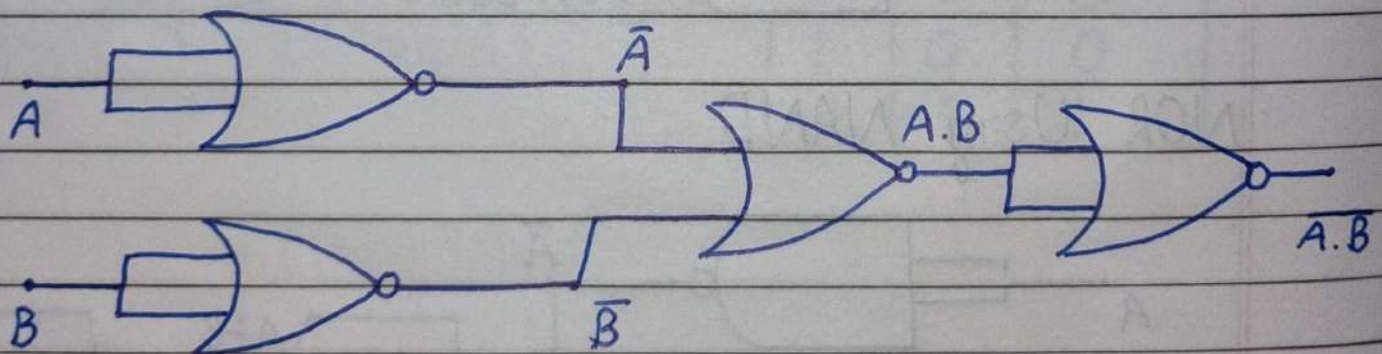
NOT Using NOR



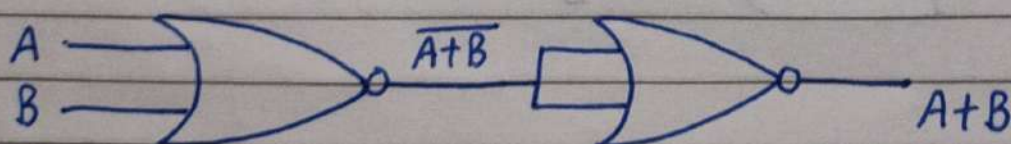
AND Using NOR



NAND Using NOR



OR Using NOR



Types of Boolean Functions

1. Sum of Product (SOP)

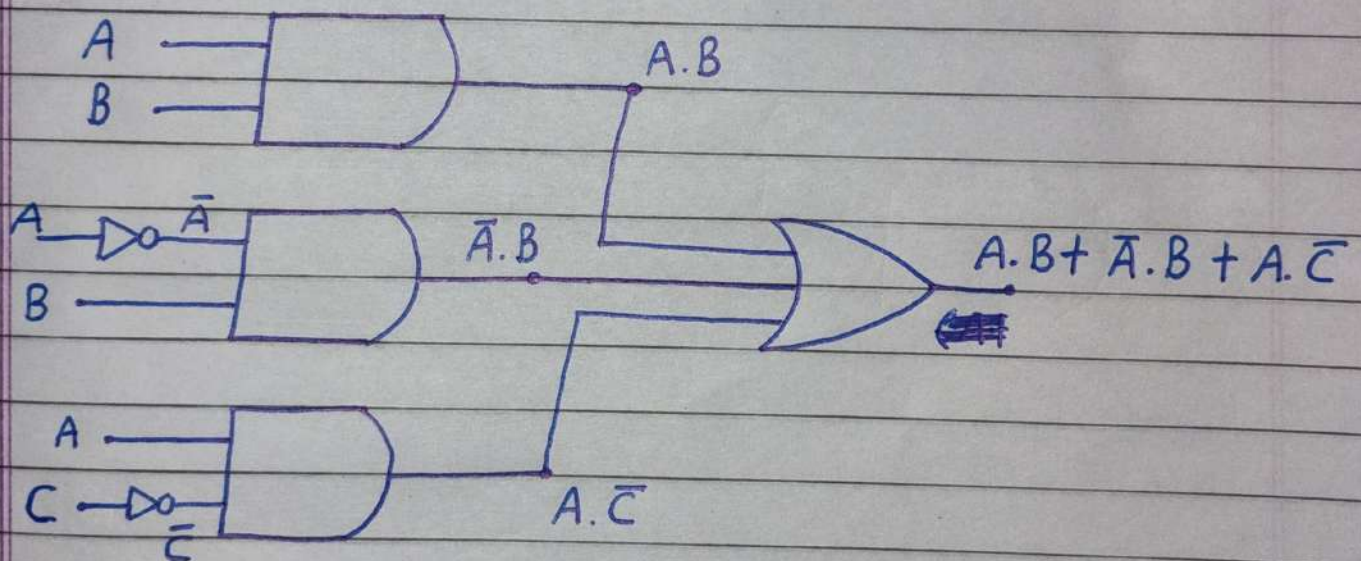
$$F = A.B + \bar{A}.B + A.\bar{C}$$

2. Product of Sum (POS)

$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

Implementation of Boolean Function Using Basic Gates

$$F = A.B + \bar{A}.B + A.\bar{C}$$



Types of Boolean Functions

1. Sum of Product (SoP)

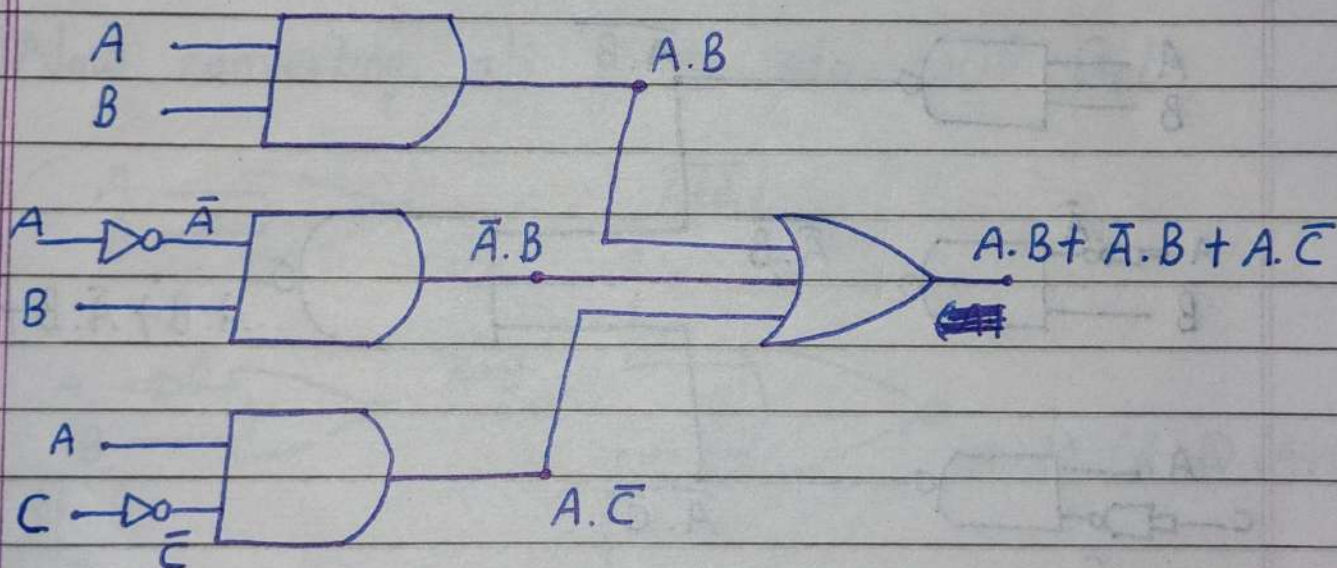
$$F = A.B + \bar{A}.B + A.\bar{C}$$

2. Product of Sum (PoS)

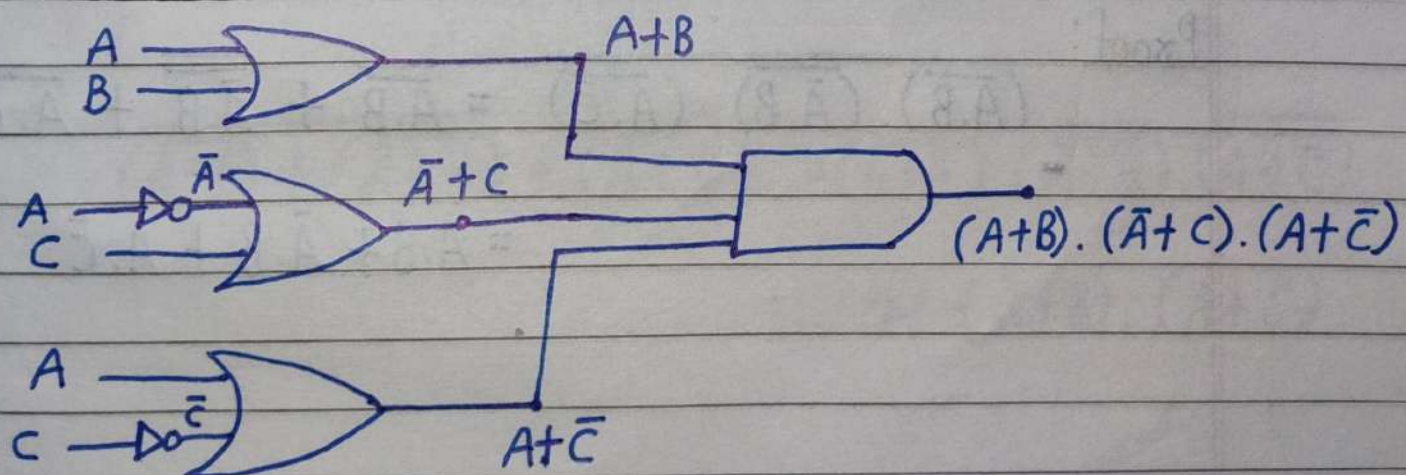
$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

Implementation of Boolean Function Using Basic Gates

$$F = A.B + \bar{A}.B + A.\bar{C}$$

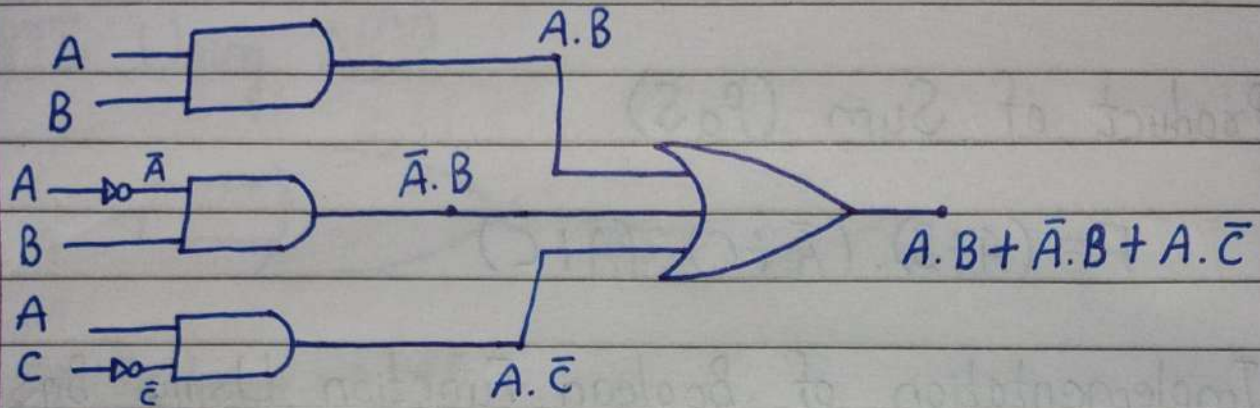


$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

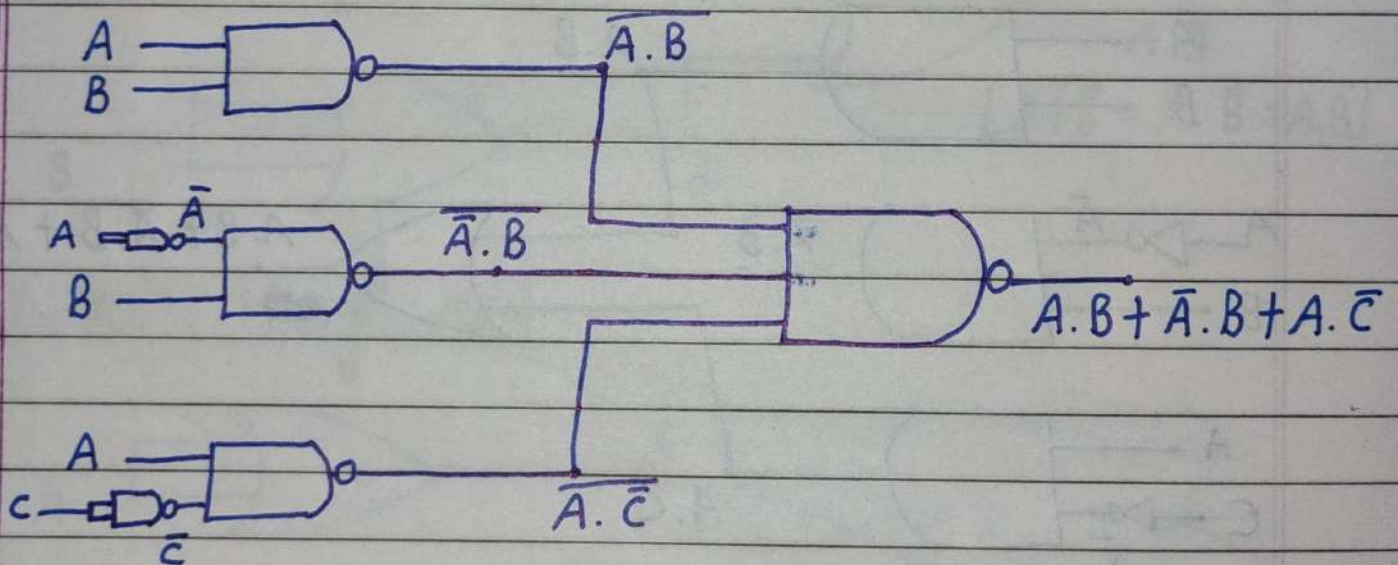


Implementation of Boolean Function Using NAND Gate Only (Universal Gate)

$$F = A.B + \bar{A}.B + A.\bar{C}$$



Now converting all gates into NAND Gate,



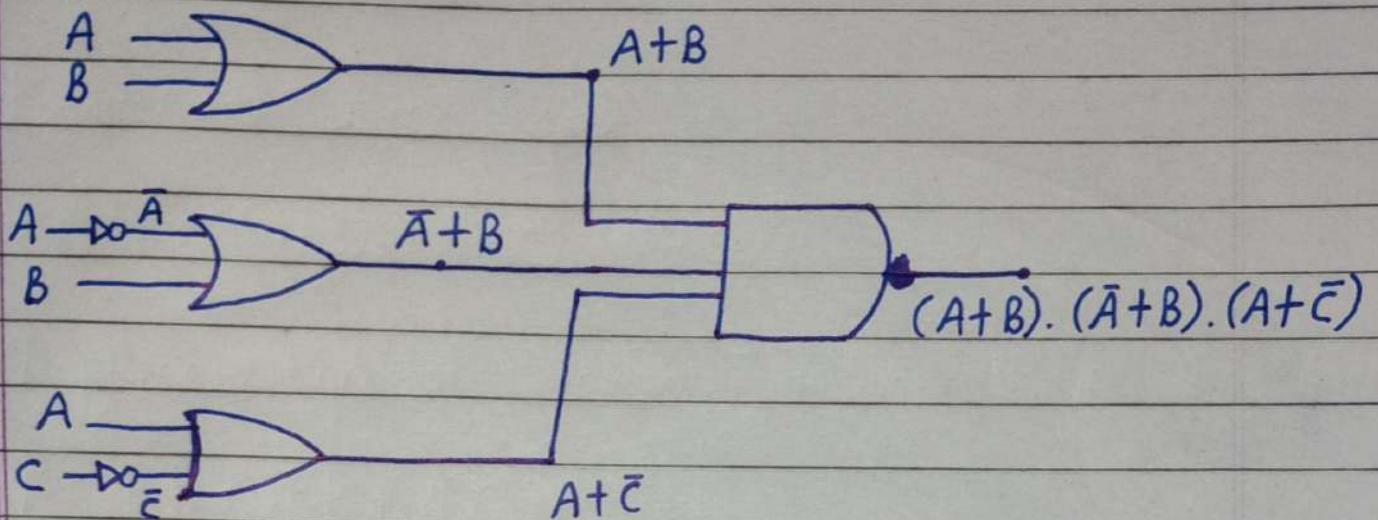
Proof:

$$\overline{(\overline{A.B})} . \overline{(\overline{\bar{A}.B})} . \overline{(\overline{A.\bar{C}})} = \overline{\overline{A.B}} + \overline{\overline{\bar{A}.B}} + \overline{\overline{A.\bar{C}}}$$

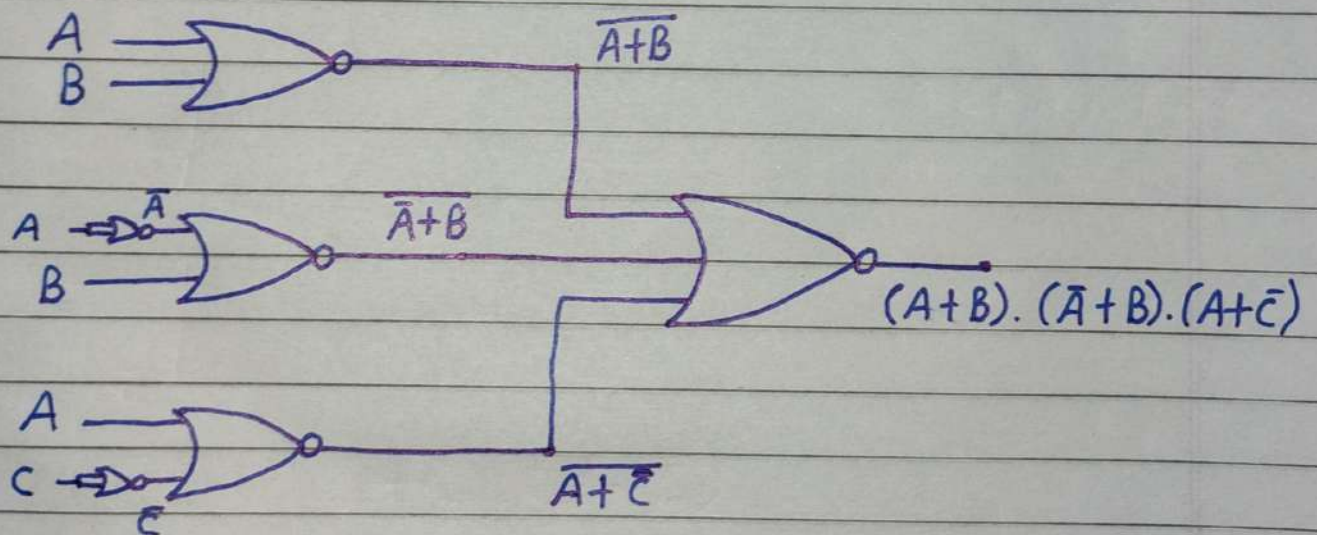
$$= A.B + \bar{A}.B + A.\bar{C}$$

Implementation of Boolean Function Using NOR Gate Only (Universal Gate)

$$F = (A+B).(\bar{A}+B).(A+\bar{C})$$



Now converting all gates into NOR Gate,



Proof:

$$\begin{aligned} \overline{(\bar{A}+B)} + \overline{(\bar{A}+B)} + \overline{(A+\bar{C})} &= \overline{(\bar{A}+B)}. \overline{(\bar{A}+B)}. \overline{(A+\bar{C})} \\ &= (A+B).(\bar{A}+B).(A+\bar{C}) \end{aligned}$$

Imp: # Binary Coded Decimal (BCD)

Decimal	BCD	Excess-3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100
10	0001 0000	0100 0011
11	0001 0001	0100 0100
:	:	:
123	0001 0010	0100 0101
	0011	0110

Binary to Gray (Cyclic) Code

$$(10101010)_2 = (\quad)_{\text{Gray}}$$

$\begin{array}{ccccccc}
& \text{XOR} & & \text{XOR} & & & \\
& \swarrow & & \swarrow & & & \\
1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\
\downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1
\end{array}$

$$(10101010)_2 = (11111111)_{\text{Gray}}$$

$$(1011001101)_2 = ()_{\text{Gray}}$$

$$\begin{array}{cccccccc} 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \end{array}$$

$$(1011001101)_2 = (1110101011)_{\text{Gray}}$$

Gray (Cyclic) Code to Binary

$$(10101010)_{\text{Gray}} = ()_2$$

$$\begin{array}{ccccccc} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ \downarrow & \swarrow \text{XOR} & \downarrow & \swarrow \text{XOR} & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}$$

$$(10101010)_{\text{Gray}} = (11001100)_2$$

$$(1011001101)_{\text{Gray}} = ()_2$$

$$\begin{array}{cccccccc} 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \end{array}$$

$$(1011001101)_{\text{Gray}} = (1101110110)_2$$

Gray (Cyclic / Unit Distance) Code

	Binary	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Boolean Algebra

Let, x be a boolean variable i.e. x is either 0 or 1.

$$x + 0 = x$$

$$x + \bar{x} = 1$$

$$x + x = x$$

$$1 + x = 1$$

$$x \cdot 1 = x$$

$$x \cdot \bar{x} = 0$$

$$x \cdot x = x$$

$$0 \cdot x = 0$$

$$\boxed{x + y \cdot z = (x + y) \cdot (x + z)}$$

Q. Find the compliment of the following functions:

$$F = A + B \cdot \bar{C}$$

$$\begin{aligned}\bar{F} &= \overline{A + B \cdot \bar{C}} = \bar{A} \cdot \overline{B \cdot \bar{C}} \\ &= \bar{A} \cdot (\bar{B} + C)\end{aligned}$$

$$\Rightarrow \bar{F} = \bar{A} \cdot (\bar{B} + C)$$

$$F = AB + (C + D) \cdot (A + \bar{C}) \cdot (\bar{A} + \bar{B})$$

$$\bar{F} = \overline{AB + (C + D) \cdot (A + \bar{C}) \cdot (\bar{A} + \bar{B})} = \bar{A} \cdot \bar{B} \cdot \overline{(C + D) \cdot (A + \bar{C}) \cdot (\bar{A} + \bar{B})}$$

$$~~= \bar{A} \cdot \bar{B}~~ = \bar{A} \cdot \bar{B} \cdot ((\bar{C} + \bar{D}) + (\bar{A} + C) + (\bar{A} + \bar{B}))$$

$$\Rightarrow \bar{F} = (\bar{A} + \bar{B}) \cdot ((\bar{C} \cdot \bar{D}) + (\bar{A} \cdot C) + (A \cdot B))$$

9. Simplify the following functions:

$$F = w\bar{x}y + wx + w\bar{y} + wx\bar{y}$$

$$F = w[\bar{x}y + x + \bar{y} + x\bar{y}]$$

$$\Rightarrow F = w[x(1 + \bar{y}) + \bar{x}y + \bar{y}]$$

$$\Rightarrow F = w[x.1 + \bar{x}y + \bar{y}] \quad [\because 1 + \bar{x} = 1]$$

$$\Rightarrow F = w[(x + \bar{x}). (x + y) + \bar{y}] \quad [\because x + yz = (x + y)(x + z)]$$

$$\Rightarrow F = w[1.(x + y) + \bar{y}] \quad [\because x + \bar{x} = 1]$$

$$\Rightarrow F = w[x + y + \bar{y}]$$

$$\Rightarrow F = w[x + 1] \quad [\because x + \bar{x} = 1]$$

$$\Rightarrow F = w.1 \quad [\because x + 1 = 1]$$

$$\Rightarrow F = w$$

$$F = \overline{A.B} + \overline{A.C}$$

$$F = \overline{A.B} . \overline{A.C}$$

$$\Rightarrow F = A.B . A.C$$

$$\Rightarrow F = A.B.C \quad [\because A.A = A]$$